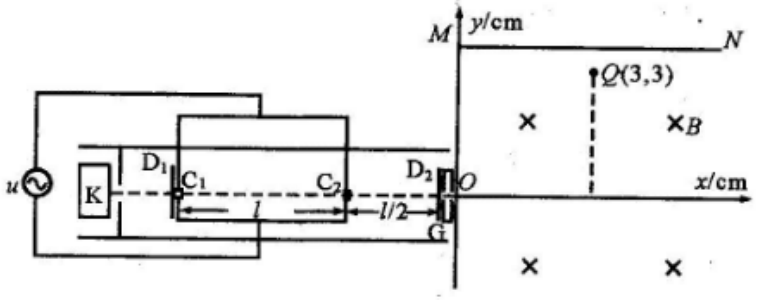
5. (25 points)

K emits positive particles continuously at various speeds, with charge q and mass m, all along the dashed line. D_1、D_2 are timed shutters; C_1、C_2 are gates; C_1 adjoins D_1; gate spacing l. Both gates carry AC voltage $u=U_0\sin(\frac{2\pi}{T}t)$ of period T. Passage occurs only instantaneously near zero voltage. Energy modulator G adjoins D_2, $l/2$ from gate C_2. At time t, G changes energy by $\Delta E_t=E_0\sin(\frac{2\pi}{T}t+\frac{\pi}{4})$, keeping velocity direction. $E_0=\frac{\sqrt{2}}{4}\times m\frac{l^2}{T^2}$. Past G, particles enter at O the uniform field right of G. Its magnitude is B, into the page; all is in vacuum. As shown, origin O defines Cartesian axes Oxy; MN is parallel to the x axis. At $t=0$ open D_1, D_2 together to admit particles to C_1. At $t=3T/4$, close D_1 to stop entry to C_1; at $t=10T/4$, close D_2 to stop entry to G. The fastest entry at O passes Q at (3cm, 3cm). For the slowest entry at O to pass Q, where must boundary MN be shifted parallel to itself in plane Oxy?



K

q

m

D_1、D_2

C_1、C_2 C_1 D_1 l

$u = U_0\sin\left(\frac{2\pi}{T}t\right)$ T

G D_2 C_2

$l/2$ t G

$\Delta E_t = E_0\sin\left(\frac{2\pi}{T}t+\frac{\pi}{4}\right)$ $E_0 = \frac{\sqrt{2}}{4}\times$